FREE GUIDE

How to read a certificate of analysis

A certificate of analysis (COA) is the one document that tells you what is actually in a laboratory vial. This guide explains what each part of it means, what it can and cannot prove, and how to tell a genuine certificate from a decorative one in about sixty seconds.

What a certificate is

A COA is a test report on one production batch (also called a lot). A laboratory takes a sample from that batch, runs a defined set of analytical tests, and reports the results against a specification. That is all it is.

It is not a licence, an approval, a registration or a statement that a material is fit for any purpose. It does not describe other batches, and it does not describe what happened to the material after the sample was taken (storage, shipping, repackaging).

> **The single most important idea**
> A certificate only means something when it is for the same lot as the vial in your hand. A "representative" certificate reused across many batches tells you about a sample someone once tested — not about the material you received.

What a complete certificate contains

Section	What it should show
Material	The compound name, and ideally its sequence or molecular formula and expected molecular weight.
Lot / batch number	A unique identifier that also appears on the vial label.
Dates	Manufacture date and test (or report) date. The test date should be after manufacture.
Laboratory	Name and address of the laboratory that ran the tests, and who authorised the report.
Tests and methods	Each test performed, the method used (e.g. HPLC, LC-MS) and the specification it was judged against.
Results	An actual measured value for each test, not just "pass" — plus the raw chromatogram or spectrum where possible.

THE THREE QUESTIONS A CERTIFICATE ANSWERS

Identity, purity and content

Most confusion comes from treating these as one number. They are three different questions, answered by different tests, and a good certificate answers all three.

1. **Identity — "Is it the right molecule?"**
   Usually answered by mass spectrometry (MS, LC-MS, ESI-MS or MALDI-TOF). The instrument measures the mass of the molecule and the report compares the observed mass with the expected (theoretical) mass. They should agree closely — typically within about one dalton for a peptide of this size, depending on the method. Without an identity result, nothing else on the page tells you which compound is in the vial.

2. **Purity — "How clean is it?"**
   Usually answered by HPLC (high-performance liquid chromatography) with UV detection. The sample is separated into its components; the main peak's share of the total peak area is reported as a percentage. 98% means the main component accounts for 98% of what the detector saw. It does not say what that main component is — that is identity's job.

3. **Content — "How much is there?"**
   A vial labelled 10 mg contains powder that is not all target molecule: freeze-dried peptides also hold water and counter-ions (often acetate or TFA salts). Net peptide content — by amino-acid analysis, nitrogen analysis or quantitative HPLC — reports the actual amount. Many certificates omit it; when it is present, it is a sign of a more thorough laboratory.

Other tests you may see

Test	What it tells you
Appearance	Colour and form (e.g. "white lyophilised powder"). A basic check that the sample looks as it should.
Water content (Karl Fischer)	How much moisture the powder holds. High water content shortens shelf life.
Counter-ion (TFA / acetate)	Residue from synthesis and purification. Reported because it affects net content.
Endotoxin / bioburden	Microbiological tests. Relevant to some laboratory applications; not always performed.
Residual solvents	Traces of solvents used during manufacture, against a stated limit.

A useful rule of thumb: purity without identity is not evidence. A vial of the wrong compound can be 99% pure. Read the identity line first, then purity, then content.

THE SIXTY-SECOND CHECK

Four tells of a genuine certificate

You do not need a laboratory to judge a certificate. These four checks catch the large majority of problems.

1. **The lot number matches the label**
   Find the lot number on the vial and on the certificate. They must be identical. If the label has no lot number, no certificate can be linked to it — which is a problem in itself.

2. **A real, named laboratory issued it**
   There should be a laboratory name, an address and an authorising signatory. Look the laboratory up: does it exist, and does it actually offer the tests listed? Being independent of the seller is a strong positive.

3. **There are actual numbers — and raw data**
   Each test should report a measured value (e.g. “observed mass 1234.6 Da against 1234.4 Da expected”, “purity 99.1%”), not only “conforms”. A chromatogram or mass spectrum attached to the report is hard to fake convincingly and easy to look at.

4. **The dates make sense**
   The test date should come after the manufacture date and be reasonably recent relative to the batch. A certificate dated years before the material was supposedly made is a mismatch worth asking about.

Going one step further

Some laboratories let you verify a report directly — by a code, a QR link to their own website, or by email. Where that is offered, use it: a verification that comes from the laboratory, not the seller, is the strongest evidence a certificate is genuine and unaltered.

If the seller says “independent testing”, it is reasonable to ask which laboratory, and whether you can see the full report for your specific lot before you order.

WHAT SHOULD WORRY YOU

Red flags, ranked

Not every imperfection means something is wrong. This list is ordered from most to least serious.

Red flag	Why it matters
No identity test at all	Without mass spectrometry (or equivalent), there is no evidence of which compound is in the vial.
Same certificate for every batch	A generic or “representative” certificate cannot describe your lot.
No lot number, or it does not match	Nothing links the report to the material you hold.
Laboratory cannot be identified	An unnamed, untraceable or seller-owned “lab” provides little independent assurance.
Results are “pass” only	No measured values means nothing to check and nothing to compare between batches.
Edited or low-quality images	Mismatched fonts, blurred numbers or cropped headers can indicate an altered document.
Purity is always exactly 99%+	Real results vary a little from batch to batch. Identical figures every time deserve a question.
No content or water data	Common, not alarming — but a more complete report is a better report.

Your one-page checklist

- ☐ Lot number on the certificate matches the vial label.
- ☐ Identity confirmed by mass spectrometry, with observed and expected mass.
- ☐ Purity by HPLC with a measured percentage (and ideally the chromatogram).
- ☐ Named, traceable laboratory — ideally independent of the seller.
- ☐ Manufacture and test dates present, in a sensible order.
- ☐ Verification route offered by the laboratory itself, where available.

About Pepco Lab. Every lot we supply is tested by an independent laboratory, and its certificate is published by lot number so it can be checked against the label. Questions about any certificate — ours or anyone else’s — are welcome at hello@pepcolab.com.